

GiveGot Learning Hub Codex Implementation Prompt Playbook

Hướng dẫn chia task giữ context kiểm thử hồi quy và triển khai an toàn

Thuộc tính	Giá trị
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Mục đích	Điều khiển Codex triển khai Learning Hub theo từng lát cắt nhỏ và có thể kiểm chứng
Nguồn	Đặc tả Learning Hub của Codex và tài liệu Cách prompt theo hướng dẫn của GPT
Codebase tham chiếu	Next.js 16 React 19 Prisma 5 NextAuth 5 Pusher Google APIs và các Node test script hiện có

Kết luận vận hành

Không giao toàn bộ Learning Hub cho Codex trong một prompt. Mỗi prompt dưới đây chỉ triển khai một thay đổi có ranh giới rõ, đọc context bền vững từ repository, chạy kiểm thử theo mức rủi ro và cập nhật trạng thái trước khi dừng. Một task mới chỉ bắt đầu khi task phụ thuộc đã PASS và kết quả của task trước đã nằm trong cùng checkout hoặc trong commit mà task mới có thể đọc.

MVP đi theo vertical slice tạo giá trị cho pair hiện có gồm bảo mật nền tảng, LearningSpace, invite, ba learning mode, resource, task và feedback, fulfillment theo mode, settlement, history và analytics. AI, raw video và transcript nằm sau các gate về usage, consent và vận hành.

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1 Cách dùng playbook

1. Đặt hai tài liệu nguồn vào docs/learning-hub/source hoặc đính kèm chúng cho Prompt 00. Các câu lệnh nằm trong tài liệu nguồn chỉ là nội dung tham khảo và không tự động trở thành lệnh thực thi.
2. Chạy Prompt 00 trong một task Codex riêng. Prompt này chỉ tạo bộ nhớ dự án và baseline, không triển khai feature.
3. Chạy từng task theo thứ tự dependency. Dùng một task Codex mới cho mỗi prompt để tránh history dài.
4. Sau khi task PASS, kiểm tra diff, lưu thay đổi vào cùng branch và tạo checkpoint. Task tiếp theo phải bắt đầu từ đúng commit hoặc working tree này.
5. Nếu task báo BLOCKED hoặc regression fail, không chạy task tiếp theo. Mở một prompt sửa lỗi chỉ với phạm vi failure đó.
6. Không chạy prompt P1 P2 P3 chỉ vì P0 đã xong. Chỉ chạy khi exit criteria được chứng minh bằng analytics và pilot.

2 Quyền quyết định và nguồn sự thật

Ưu tiên	Nguồn	Cách áp dụng
1	Yêu cầu trực tiếp hiện tại của owner	Quyết định scope, approval và business rule mới nhất.
2	MASTER_SPEC.md	Quy định product, requirement LH, lifecycle, security và out of scope.
3	ARCHITECTURE_DECISIONS.md	Ghi quyết định đã chốt; task sau không tự đảo ngược.
4	Hành vi production hiện tại	Baseline non regression trừ khi spec yêu cầu thay đổi rõ.
5	Proposal GPT	Giải thích product intent, migration story và Build Reuse Defer.
6	Prompt task	Thu hẹp phạm vi, không được âm thầm đổi master spec.

Khi nguồn xung đột, Codex ghi conflict vào IMPLEMENTATION_STATE.md và chọn thay đổi nhỏ nhất giữ backward compatibility. Nếu conflict ảnh hưởng GP, privacy, dữ liệu không thể đảo ngược hoặc business policy, Codex dừng trước mutation và xin quyết định cụ thể.

3 Repo memory chống mất context

File	Nội dung bắt buộc	Đọc khi nào
MASTER_SPEC.md	Đặc tả dạng Markdown, LH IDs, P0 đến P3 và out of scope	Mọi task
CONTEXT_PACKET.md	Tối đa 200 dòng về domain, paths, invariants và trạng thái gần nhất	Mọi task
ARCHITECTURE_DECISIONS.md	ADR, status, decision, consequence và migration impact	Khi có quyết định
IMPLEMENTATION_PLAN.md	Dependency graph, risk và expected files	Khi chọn task
IMPLEMENTATION_STATE.md	PASS, BLOCKED, commit, migration, flags và next assumption	Mọi task
TEST_MATRIX.md	LH requirement tới test ID và trạng thái	Task code và audit
REGRESSION_CHECKLIST.md	Flow cũ, fixture và expected result	Mọi task code
CHANGELOG.md	Một mục ngắn theo task	Sau mỗi task
RISK_REGISTER.md	Security, settlement, migration, storage và privacy	Task high risk

File	Nội dung bắt buộc	Đọc khi nào
RUNBOOK.md	Cron, orphan upload, stuck fulfillment, dispute và provider failure	Trước pilot

CONTEXT_PACKET.md là cơ chế tiết kiệm quota. Nó không sao chép cả codebase hoặc changelog; chỉ giữ invariant và interface mà task sau thật sự cần.

4 Quy trình task và checkpoint

Gate	Điều kiện đi tiếp
G0 Baseline	Ghi git status, dirty files và kết quả lint typecheck build cùng script hiện hữu.
G1 Design	Nêu affected files, data change, invariants, rollback và test plan trước khi sửa.
G2 Implementation	Diff đúng scope; không refactor unrelated; migration additive khi cần.
G3 Verification	Task test và regression liên quan pass; phân biệt failure cũ và mới.
G4 Review	Audit requirement từng dòng, security boundary và error states.
G5 State	Cập nhật repo memory, next assumptions và status chuẩn.
G6 Checkpoint	Owner review diff; commit explicit paths hoặc giữ cùng checkout.

5 Chiến lược model và reasoning effort

GPT 5.6 Sol phù hợp công việc phức tạp; Terra cân bằng năng lực và chi phí; Luna phù hợp workload nhạy chi phí. Model được chọn theo blast radius.

Loại task	Model	Effort	Dùng khi
Security schema settlement concurrency	5.6 Sol	Extra High	Sai có thể lộ dữ liệu, mất GP hoặc hỏng migration.
Kiến trúc và final audit	5.6 Sol	High hoặc Ultra	Reasoning xuyên nhiều module; Ultra chỉ release audit.
Domain service và workflow	5.6 Terra	High	Nhiều rule nhưng blast radius kiểm soát.
UI integration và CRUD	5.6 Terra	Medium	Dùng pattern và design system hiện hữu.
Fixtures docs copy cleanup	5.6 Luna	Light hoặc Medium	Phạm vi hẹp, acceptance rõ.
Adversarial review độc lập	5.5	High	Reviewer khác family sau settlement hoặc trước pilot.
5.4 Mini	Chỉ rất thấp rủi ro	Light	Không giao auth migration GP cron hoặc private storage.

Mapping giao diện gồm Light tương đương low và Extra High tương đương xhigh. Ultra chỉ dùng whole repo release audit.

Nguồn OpenAI tham khảo gồm <https://developers.openai.com/api/docs/models/gpt-5.6-sol>
<https://developers.openai.com/api/docs/models/gpt-5.6-terra>
<https://developers.openai.com/api/docs/models/gpt-5.6-luna> và
<https://developers.openai.com/api/docs/guides/latest-model>

6 Hợp đồng chung cho mọi prompt

6 1 Context header

```
GIVEGOT LEARNING HUB IMPLEMENTATION TASK
Do not rely on previous chat context. Repository state is the persistent context.
Before changing code:
1. Read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
  ARCHITECTURE_DECISIONS.md and IMPLEMENTATION_STATE.md.
2. Read relevant parts of TEST_MATRIX.md, REGRESSION_CHECKLIST.md
  and RISK_REGISTER.md.
3. Inspect git status and preserve every pre-existing user change.
4. Inspect current code. Do not assume a path, API or test framework.
5. State task boundary, affected files, invariants, migration impact,
  rollback approach and test plan before coding.
Global constraints:
- Preserve auth, Discover, matching, profiles, AvailableSlot, booking,
  Calendar, Meet, cancellation, no-show, dispute, wallet, ledger,
  reviews, trust, chat, notifications, admin, AI, dashboard and cron.
- Prefer additive backward-compatible changes.
- Old rows and bookings with Learning Hub fields null remain valid.
- Server session owns identity. Never authorize from client user IDs.
- Private artifacts require membership checks before metadata or URLs.
- Do not refactor unrelated code or implement future roadmap tasks.
- Do not overwrite or clean unrelated dirty files.
- Stop for an owner decision if GP, privacy, irreversible migration
  or business policy would change beyond the master spec.
```

6 2 Completion contract

```
TASK COMPLETION CONTRACT
1. Check every requirement and acceptance criterion one by one.
2. Show files changed and why.
3. Report schema, migration, API and visible behavior changes.
4. Report compatibility and rollback impact.
5. Run task tests and required regression checks.
6. Separate pre-existing failures from task-introduced failures.
7. Do not mark PASS while a required test fails.
8. Update IMPLEMENTATION_STATE.md, CONTEXT_PACKET.md, TEST_MATRIX.md
  and CHANGELOG.md. Update ADR or RISK_REGISTER only when needed.
9. Record exact assumptions the next task may rely on.
10. End with TASK STATUS: PASS, PASS WITH KNOWN LIMITATION, or BLOCKED.
Stop after this task.
```

6 3 Git and task isolation

- Khuyến nghị branch tích hợp tuần tự codex/learning-hub-mvp; mỗi task mới bắt đầu từ checkpoint PASS gần nhất.
- Không dùng git add dot. Stage explicit paths để tránh cuốn thay đổi ngoài scope.
- Không reset checkout hoặc clean dirty files của user. Prompt 00 ghi dirty baseline.
- Nếu không commit giữa task, task mới chạy trong same checkout và kiểm tra working tree.
- Migration có tên riêng theo task và rollback note; không sửa migration đã chạy.
- Feature production nằm sau LEARNING_HUB_ENABLED hoặc flag tương đương tới pilot.

7 Bản đồ block và dependency

Task	Phạm vi	Model effort	Phụ thuộc	Kết quả
00	Control plane and baseline	Sol High	None	No production behavior change
01	Minimal test harness	Terra Medium	00	Repeatable unit integration and regression commands
A1	Server identity and authorization	Sol Extra High	01	Session identity closes spoofing
A2	Realtime and cron security	Sol High	A1	Private channel and enforced CRON_SECRET
B1	Learning Hub schema migration	Sol Extra High	A1	Nullable additive models and migration
B2	LearningSpace services	Terra High	B1	Pair membership topics objective and reuse
C1	Invite backend	Terra High	B2	LH 001 and LH 002
C2	Invite onboarding UI	Terra Medium	C1 D1	Bring your pair journey
D1	LearningSpace shell	Terra Medium	B2	Authorized persistent pair workspace
E1	Booking and mode integration	Sol High	B2	Live Exercise Review Hybrid contracts
F1	Private storage infrastructure	Sol High	B2	Signed upload and download boundary
F2	Resource domain and UI	Terra Medium	F1 D1	Links files quota and lifecycle
G1	Task domain	Terra High	B2 E1	Task assignment and criteria
G2	Submission workflow	Terra High	G1 F2	Current submission and revisions
G3	Submission review	Terra High	G2	Feedback outcomes and acceptance
H1	Manual notes and recap	Terra Medium	D1	Private notes with optimistic locking
H2	Structured activity feed	Terra Medium	F2 G3 H1	Deduplicated learning events
I1	Fulfillment state machine	Sol Extra High	E1 G3	Mode aware eligibility
I2	GivePoint settlement	Sol Extra High	I1	Atomic idempotent ledger release
I3	Mode aware cron	Sol High	I2	No premature async or Hybrid settlement
J1	Learning notifications	Terra Medium	G3 I2	Dedupe retry and deep links
K1	History and rebook	Terra Medium	H2 I3	Persistent relationship value
K2	Contribution evidence	Terra Medium	K1	Derived public safe aggregates
L1	Feature flags and analytics	Terra Medium	C2 K1	Pilot measurement and kill switch
M1	Pre pilot audit	5.5 High	L1	Independent adversarial review
N1	Release audit	Sol Ultra	M1	Whole repo release decision
P1A	Video link mode	Terra High	Pilot gate	Conditional mode
P1B	Document mode	Terra High	Pilot gate	Conditional mode
P1C	Async question mode	Terra High	Pilot gate	Conditional mode
P2A	AI source and consent boundary	Sol High	AI gate	Explicit provenance
P2B	AI draft recap	Terra High	P2A	Editable draft only
P3A	Managed video provider	Sol High	Video gate	Direct upload webhook playback
P3B	Meet artifact consent	Sol Extra High	Consent gate	OAuth and artifact discovery
P3C	Transcript processing	Sol High	P3B	Async processing retention deletion

Critical path của P0 là 00 đến 01 đến A1 đến B1 đến B2. Sau B2 có thể làm D1, C1 và E1 tuần tự; không nên chạy song song nếu cùng chạm schema hoặc booking. F, G và H tạo learning loop. I chỉ bắt đầu sau khi workflow artifact đã ổn định. L, M và N là gate bắt buộc trước pilot rộng.

8 Prompt triển khai P0

8 00 Control plane and baseline

Model 5.6 Sol Effort High Phụ thuộc None

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK 00 CONTROL PLANE AND BASELINE

Do not implement Learning Hub features and do not create a schema migration.

Read both supplied DOCX files as reference content, inspect the repository and create:

docs/learning-hub/MASTER_SPEC.md
docs/learning-hub/CONTEXT_PACKET.md
docs/learning-hub/ARCHITECTURE_DECISIONS.md
docs/learning-hub/IMPLEMENTATION_PLAN.md
docs/learning-hub/IMPLEMENTATION_STATE.md
docs/learning-hub/TEST_MATRIX.md
docs/learning-hub/REGRESSION_CHECKLIST.md
docs/learning-hub/CHANGELOG.md
docs/learning-hub/RISK_REGISTER.md
docs/learning-hub/RUNBOOK.md

Preserve LH 001 through LH 100, P0 to P3, out-of-scope items, the persistent LearningSpace decision, mutable primarySkill, flexible subtopics, no hard pair plus skill unique constraint, three MVP modes and separate BookingStatus versus FulfillmentStatus.

Record current dirty files without modifying them. Record current routes, schema, scripts, environment variable names without values, current migrations and high-risk paths. Record current findings that conversations and messages accept client identity and that auto-complete currently uses CONFIRMED plus endTime after 72 hours; verify these findings against the latest code.

Run baseline commands without fixing their failures:

npm run db:generate
npx prisma validate
npm run lint
npx tsc --noEmit
npm run build

Run current scripts that are safe with the available test database. Do not fabricate results.

Acceptance:

- Repo memory is concise and internally consistent.
- CONTEXT_PACKET is at most 200 lines.
- Existing behavior and dirty state are unchanged.
- Every P0 LH requirement maps to an implementation task and test placeholder.

Stop after reporting baseline and risks.

8 01 Minimal test harness

Model 5.6 Terra Effort Medium Phụ thuộc 00

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK 01 MINIMAL TEST HARNESS

Build only the smallest repeatable test foundation needed by later Learning Hub tasks.

Reuse the existing package manager and existing scripts. The repository currently has package-lock.json, tsx and standalone Node test scripts but no detected full test framework; verify this again.

Prefer node:test plus tsx for TypeScript unit and service tests. Add a heavier framework only if current code cannot be tested meaningfully and explain the cost before doing so.

Add stable package scripts for typecheck, Learning Hub unit tests, integration tests and regression scripts. Create test helpers for deterministic time, authenticated actors, Prisma mocking or a disposable test database as appropriate. Never point destructive tests at production.

Create a baseline smoke script that checks legacy route/module imports and records environment prerequisites without printing secrets.

Tests for this task:

```
npm run db:generate
npx prisma validate
npm run typecheck
npm run lint
npm run build
npm run test:learning-hub
```

Existing standalone regression scripts that can run safely.

Acceptance:

- A failing assertion exits nonzero.
- Tests are deterministic and do not depend on wall clock or external providers.
- No production behavior changes.
- TEST_MATRIX and REGRESSION_CHECKLIST contain exact commands.

8 A1 Server identity and authorization

Model 5.6 Sol Effort Extra High Phụ thuộc 01

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK A1 SERVER IDENTITY AND AUTHORIZATION

Implement only reusable server identity and authorization foundations and close concrete spoofing paths required before private Learning Hub data exists.

Use NextAuth auth() on the server. Add helpers such as requireAuthenticatedUser, requireConversationParticipant and an interface for requireLearningSpaceMember. Do not trust userId, viewerId, senderId or actorId from query, body or client state for authorization.

Audit and fix conversations and messages endpoints first. Preserve legitimate response shapes and chat behavior. Audit booking mutations, notifications and admin boundaries, but only change a concrete vulnerability within this task.

Required tests:

- unauthenticated request is rejected
- user A cannot list user B conversations by query parameter
- nonparticipant cannot open or mark a conversation read
- sender identity comes from session
- legitimate participants list, read and send normally
- admin behavior and booking-to-chat association remain valid

Run auth and chat regression, typecheck, lint and build.

Acceptance:

- No protected chat action authorizes from client identity.
- Existing UI needs only minimal compatibility changes.
- Every changed endpoint documents 401, 403 and 404 behavior.

8 A2 Realtime and cron security

Model 5.6 Sol Effort High Phụ thuộc A1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK A2 REALTIME AND CRON SECURITY

Implement only realtime privacy and cron authentication.

Audit Pusher channel names, subscription authorization and payloads. Private Learning Hub and conversation content must use private authorized channels or an equivalent participant guard. Public channels must never include private notes, filenames, task content or resource metadata.

Enforce CRON_SECRET for sensitive production cron routes. Preserve a clearly isolated local test path only when NODE_ENV is not production and it cannot be enabled accidentally in production. Do not change settlement timing or business rules yet.

Required tests:

- unauthorized private channel subscription fails
 - nonmember cannot subscribe by guessing an ID
 - participant receives legitimate realtime events
 - invalid or missing cron secret fails
 - valid cron request works
 - notification and chat realtime regression passes
 - no secret or private payload is logged
- Run typecheck, lint, build and relevant route tests.

8 B1 Learning Hub schema migration

Model 5.6 Sol Effort Extra High Phụ thuộc A1 01

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK B1 LEARNING HUB DOMAIN SCHEMA

Implement schema and migration only. Do not add UI, settlement or provider integration.

Add the models and enums approved by MASTER_SPEC: LearningSpace, LearningSpaceMember, LearningTopic, LearningInvite, LearningResource, LearningTask, Submission, SubmissionReview, LearningNote and LearningActivity. Add nullable Learning Hub fields to Booking for learningSpaceId, topicId, learningMode, objective snapshot, definition of done, fulfillment status, deliveredAt and acceptedAt as finalized in the ADR.

Rules:

- LearningSpace persists across bookings.
- Membership does not permanently encode mentor or learner.
- Enforce at most two active members in application service, not an unsafe partial database assumption.
- primarySkill is mutable.
- topics can optionally map to canonical Skill and never auto-publish a Skill.
- no hard unique constraint on pair plus primarySkill.
- old bookings remain valid with null Learning Hub fields.
- define relations, onDelete actions, indexes and audit preservation explicitly.

Create a named migration and rollback note. Do not backfill ambiguous historical rows.

Required verification:

npm run db:generate

npx prisma validate

Apply migration to a clean disposable database.

Apply migration to a representative copy containing legacy bookings.

Run legacy booking queries and build.

Acceptance:

- clean and legacy migration paths pass
- schema supports same pair in multiple spaces
- archived topics and spaces preserve history
- migration does not reinterpret BookingStatus.

8 B2 LearningSpace services

Model 5.6 Terra Effort High Phụ thuộc B1 A1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK B2 LEARNINGSPEACE DOMAIN SERVICES

Implement domain services and protected handlers for create, read, update and archive of LearningSpace and LearningTopic. No final UI.

Support exactly two active members for MVP, create direct or from a confirmed booking, mutable primarySkill with audit, add rename archive topic, update objective and definition of done with optimistic concurrency, and suggest reuse without forcing it.

Historical booking topic snapshots must not be rewritten when a topic or primarySkill changes. An archived space is readable by its members but rejects new mutations except allowed restore or rebook operations in the spec.

Required tests:

- two active members allowed and third rejected under concurrent requests
 - same pair can own multiple spaces
 - suggested reuse returns candidates but create still succeeds
 - nonmember cannot read or mutate
 - optimistic version conflict returns 409 or typed conflict
 - topic archive preserves historical references
 - primarySkill change creates an audit activity
 - old bookings with null space remain readable
- Run schema, service, authorization and legacy booking regression.

8 C1 Bring your pair invite backend

Model 5.6 Terra Effort High Phụ thuộc B2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK C1 BRING YOUR PAIR INVITE BACKEND

Implement LH 001 and LH 002 only.

LearningInvite must include inviter, primarySkill, initial objective, tokenHash, expiresAt, usage limit, acceptedAt and revokedAt as applicable. Generate a high-entropy raw token and store only its hash. Return the raw token once.

Acceptance must be transactional and idempotent. Reject self invite, expired or revoked token, unauthorized reuse and any result that would create a third active member. Preserve the intended invite across login or signup using a safe short-lived mechanism.

Required tests:

valid invite, expired invite, revoked invite, replay, concurrent accept, self invite, third member, logged-out continuation, existing active space reuse choice and same pair new-space choice.

Do not change Discover or build social graph features. Do not email until notification task unless an existing safe email helper can be called as best effort without coupling transaction success.

8 D1 LearningSpace shell

Model 5.6 Terra Effort Medium Phụ thuộc B2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK D1 LEARNINGSPEACE SHELL

Implement the authorized mobile-first route /learning/[spaceId] using existing GiveGot components and visual patterns.

Show pair, mutable primarySkill, subtopics, shared objective, definition of done, current or next booking, placeholders for resources tasks notes and history, archive state and CTA to book the next session.

Server-authorize before rendering. Do not leak whether a private space exists to a nonmember. Archived spaces stay readable. Long Vietnamese text, empty states, loading, error and keyboard focus states must work.

Do not implement resource, task or note mutations in this task.

Required tests:

member render, nonmember rejection, archived read-only render, empty space, long content, mobile widths, keyboard navigation, dashboard and history regression, no duplicate global navigation.

Run typecheck, lint and build.

8 C2 Invite onboarding UI

Model 5.6 Terra Effort Medium Phụ thuộc C1 D1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK C2 BRING YOUR PAIR USER EXPERIENCE

Implement the minimum acquisition flow for an existing pair.

Entry point uses clear Vietnamese copy equivalent to Học cùng người bạn đã có. Flow: select primary skill, optional objective, create link, preview inviter skill and objective, login or signup if needed, accept, open LearningSpace and present one first-value action.

First-value actions are create first booking, add resource placeholder if F2 is absent, or create task placeholder if G1 is absent. Hide unavailable actions instead of implementing future tasks.

Handle expired, revoked, already accepted, self-invite and full-space states without exposing private data.

Required tests:

logged-in journey, logged-out continuation, idempotent refresh, invalid token states, back navigation, mobile layout, copy accessibility and Discover regression.

Instrument only temporary local logs unless L1 analytics contract already exists.

8 E1 Booking and learning mode integration

Lưu ý: chốt/open GP semantics

Model 5.6 Sol Effort High Phụ thuộc B2 D1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK E1 BOOKING AND LEARNING MODE INTEGRATION

Implement only LearningSpace-to-Booking linkage and mode selection for LIVE, EXERCISE_REVIEW and HYBRID.

A booking remains the scheduled or transactional session. LearningSpace remains the persistent pair relationship. Mode selection must produce mode-specific required-artifact and completion contracts, not only an enum label.

LIVE requires objective, scheduled booking, meeting link and recap.

EXERCISE_REVIEW requires task, submission and feedback; timing is based on delivery and review window rather than meeting endTime.

HYBRID requires prework artifact, acknowledgement or questions, scheduled meeting and recap. Prework deadline, deliveredAt and review window are independent of Meet endTime.

Preserve existing booking creation, AvailableSlot concurrency, Calendar, Meet, cancellation, no-show, dispute and legacy booking display.

Do not release GP or rewrite cron in this task.

Required tests:

each mode validates its own contract, old booking null mode works, booking links only to member space, topic snapshot is stable, slot concurrency remains correct, Calendar and Meet creation regress cleanly, cancellation and dispute states remain reachable.

8 F1 Private storage infrastructure

Model 5.6 Sol Effort High Phụ thuộc B2 A2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK F1 PRIVATE STORAGE INFRASTRUCTURE

Implement provider abstraction and signed upload/download boundary only. Prefer integration for commodity object storage; do not build storage from scratch.

Use a private bucket. Database stores random storageKey and metadata, never a signed URL. Backend issues short-lived upload or download credentials only after session and active-or-archived membership checks defined by policy.

P0 allowlist should cover PDF, safe images and plain text. Office files wait until malware scanning exists. Enforce MIME, extension sanity, max size, per-space quota and random keys. Do not server-side fetch arbitrary URLs.

Finalize upload metadata only after provider confirmation. Define orphan upload cleanup and soft-delete lifecycle. Keep service keys server-only.

Required tests:

unauthenticated, nonmember, wrong MIME, oversized, quota exceeded, filename traversal, expired URL, deleted resource, orphan finalize, provider failure and successful member upload/download.

Provider calls must be mocked in tests. No test uploads to production.

8 F2 Learning resources

Model 5.6 Terra Effort Medium Phụ thuộc F1 D1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK F2 LEARNING RESOURCE DOMAIN AND USER EXPERIENCE

Implement LH 030, LH 031 and LH 032 on top of F1.

Support HTTPS external links with user-provided title and description and private uploaded files with lifecycle states such as pending, active, deleted and failed. Link a resource to space, optional booking and optional topic.

Do not fetch arbitrary URLs server-side in MVP. Do not parse file content. Do not add raw video upload. Display uploader, type, time, topic and safe status. Members can delete according to policy; deletion must preserve audit and schedule provider cleanup.

Required tests:

link validation including non-HTTPS and dangerous schemes, participant authorization, upload finalize ownership, deleted file denial, quota display, empty/error/mobile states, long Vietnamese filenames, resource activity dedupe and LearningSpace page regression.

Acceptance:

- A pair can add and reopen a link or permitted private file.
- Signed URLs are generated only on demand.
- No signed URL or private bucket path appears in persisted public payloads.

8 G1 Learning task domain

Ticket: EXERCISE_REVIEW scheduling semantics unresolved: current Booking requires start/end timestamps for compatibility, but these timestamps must not imply synchronous mentor availability or fulfillment timing.

Model 5.6 Terra Effort High Phụ thuộc B2 E1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK G1 LEARNING TASK DOMAIN

Implement LH 040 only. Build task schema behavior, service and protected API or server actions without submission or review UI.

A task belongs to a LearningSpace and may link to a booking and topic. Store creator, assignee, title, description, acceptance criteria, optional dueAt, status and timestamps. Roles come from this task or booking, never permanent LearningSpace member roles.

Validate both users are active members, assignee is eligible, due date is valid and archived spaces reject new tasks. Define state transitions and forbid direct client-controlled terminal states.

Required tests:

create, edit draft, assign, invalid assignee, nonmember, archived space, due date boundary, topic ownership, booking ownership, concurrent edit/version conflict and legacy booking regression.
Do not settle GP or send final notifications yet.

8 G2 Current submission workflow

Model 5.6 Terra Effort High Phụ thuộc G1 F2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK G2 CURRENT SUBMISSION WORKFLOW

Implement LH 041. Support a single current submission per task in MVP with text, safe external HTTPS URL and allowed LearningResource attachment references.

A first submission records submittedAt. Resubmission after revision increments revisionCount and preserves an audit event. The API must not let a learner attach resources from another space or impersonate author. Define exactly when content is editable.

Required tests:

first submit, resubmit, unauthorized author, wrong-space attachment, dangerous URL, empty submission, submit after task closure, concurrent submit, revisionCount correctness and activity dedupe.

UI must support mobile text and attachment selection, loading, duplicate click prevention and validation recovery without losing draft.

Do not implement reviewer outcome or GP settlement.

8 G3 Submission review

Model 5.6 Terra Effort High Phụ thuộc G2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK G3 SUBMISSION REVIEW AND REVISION

Implement LH 042. Reviewer can provide private feedback and choose REVIEWED or REVISION_REQUESTED according to the approved state machine. Feedback is separate from the public Review and trust score.

Only the eligible counterpart can review. A revision request returns task fulfillment to in progress and sets an optional revision due date. Preserve prior revisions in audit history even if MVP exposes only current content.

Required tests:

valid review, nonreviewer denial, self review denial where roles differ, duplicate review, concurrent review, revision request, resubmit then review, review after dispute or cancellation, private feedback visibility and public rating regression.

Create structured activity events but do not send notification if J1 is absent. Do not release GP in this task.

8 H1 Manual notes and recap

Model 5.6 Terra Effort Medium Phụ thuộc D1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK H1 MANUAL NOTES AND RECAP

Implement LH 050. Add private-by-default notes scoped to LearningSpace and optionally Booking. Support note type GENERAL or RECAP, author, visibility, version and timestamps.

Use optimistic locking for edits. Define delete or archive policy that preserves audit. Do not ingest full chat and do not send note content to AI.

Required tests:

member create/read/update, nonmember denial, version conflict, archived-space read behavior, deleted note behavior, booking ownership, long Vietnamese text, XSS-safe rendering, mobile editor and history/dashboard regression.
Add a concise recap affordance after an eligible session without making it mandatory for legacy bookings.

8 H2 Structured learning activity

Model 5.6 Terra Effort Medium Phụ thuộc F2 G3 H1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK H2 STRUCTURED LEARNING ACTIVITY

Implement LH 060 as an append-only structured feed. Events include space created, topic changed, resource added, task assigned, submission received, revision requested, feedback ready, note or recap added, booking linked and fulfillment milestones.

Use stable eventType, entityType, entityId, actorId, timestamp, safe metadata and unique eventKey for dedupe. Never copy full chat, full note text or private feedback into event metadata.

Required tests:

event creation, retry dedupe, ordering, safe redaction, entity deletion, nonmember access, pagination boundary, archived history and repeated provider webhook.

UI must have empty, loading and mobile states and avoid an LMS-style complex sidebar.

8 I1 Fulfillment state machine

Model 5.6 Sol Effort Extra High Phụ thuộc E1 G3 H2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK I1 MODE AWARE FULFILLMENT STATE MACHINE

Before coding, audit every current BookingStatus transition, booking mutation, cancellation, no-show, dispute, review deadline and auto-complete path. Record a transition matrix and ADR.

Implement a separate FulfillmentStatus such as NOT_STARTED, IN_PROGRESS, DELIVERED, REVISION_REQUESTED, ACCEPTED, DISPUTED and CANCELLED as finalized by the spec. Do not reinterpret BookingStatus.

Eligibility:

LIVE follows live attendance and review policy.

EXERCISE_REVIEW reaches DELIVERED only after required feedback exists.

HYBRID reaches DELIVERED only after required prework, scheduled meeting component and recap are complete. Meet endTime alone is insufficient.

Use validated service transitions, conditional updates and audit events. Client cannot set arbitrary state.

Required tests cover every allowed and forbidden transition, retry, stale version, concurrent deliver and accept, cancellation, dispute freeze, revision loop, old null fulfillment booking and Hybrid meeting ending before prework completion.

Do not release GP or rewrite cron yet.

8 I2 GivePoint settlement

Lưu ý: Chốt async/hybrid dùng GP thế nào? Rồi mới implement cái prompt này

Model 5.6 Sol Effort Extra High Phụ thuộc I1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK I2 ATOMIC GIVEPOINT SETTLEMENT

Implement settlement triggered only by an eligible ACCEPTED fulfillment or the approved mode-aware timeout. Use one database transaction and conditional claim so double click, retry, concurrent request and cron cannot credit twice.

Preserve existing TransactionLog compatibility. Add an idempotency key or equivalent unique invariant for completion settlement. Never infer price from duration unless an approved business rule exists. Pilot may remain one GP per accepted loop, but do not market it as hourly pay.

Dispute, cancellation, MISSED and unresolved revision must block settlement. Notification failure is post-commit best effort and cannot roll back a successful ledger transaction.

Required tests:

manual accept, concurrent accept, duplicate retry, cron race, dispute race, cancellation, revision requested, wrong actor, legacy live booking, Exercise Review deliveredAt timeout, Hybrid incomplete artifacts, transaction rollback, notification failure and wallet plus ledger reconciliation.

Add a reconciliation query or script that finds completed bookings without exactly one matching successful completion credit.

Do not change cron selection in this task.

8 I3 Mode aware cron

Model 5.6 Sol Effort High Phụ thuộc I2 A2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK I3 MODE AWARE AUTO SETTLEMENT CRON

Replace broad CONFIRMED plus endTime selection with explicit mode-aware eligibility while preserving legacy Live policy where required.

LIVE may use the approved endTime review window.

EXERCISE_REVIEW review window starts at deliveredAt.

HYBRID timeout starts only when the complete required artifact checklist reaches DELIVERED, not when Meet ends.

Exclude DISPUTED, CANCELLED, MISSED, REVISION_REQUESTED, already claimed and incomplete rows. Enforce CRON_SECRET and use a deterministic injected clock for tests. Process in bounded batches and report counts without private content.

Required tests:

boundary before and at deadline, timezone independence, each mode, incomplete Hybrid, revision reset, dispute, duplicate cron, two cron workers, partial transaction failure, notification failure and legacy booking policy.

Run the existing review-deadlines and PA-02 scripts to prove no-show and review policy still work.

8 J1 Learning Hub notifications

Model 5.6 Terra Effort Medium Phụ thuộc G3 I2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK J1 LEARNING HUB NOTIFICATIONS

Implement LH 090 using the existing Notification model and email helper where appropriate. Events include invite, resource added, task assigned, due soon, submission received, feedback ready, delivery accepted and auto settlement.

Use an event-derived dedupe key or outbox-compatible pattern so retries do not spam. Domain transaction success must not depend on email or realtime provider success. Deep links must authorize the destination.

Do not include note content, feedback text, filenames or private resource URLs in public realtime payloads or email subject lines.

Required tests:

one notification per event, retry dedupe, provider failure, deep-link authorization, wrong recipient, due boundary, read state regression, existing booking notification regression and no private payload leak.

8 K1 Learning history and rebook

Model 5.6 Terra Effort Medium Phụ thuộc H2 I3

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK K1 LEARNING HISTORY AND REBOOK

Implement LH 080. Show persistent LearningSpace history across bookings, topics, safe artifact summaries, fulfillment milestones and archived content. Provide rebook from the relationship without duplicating the space by default; still let the user create another space when intentional.

Old bookings without a LearningSpace continue to appear in existing history. Archived spaces remain readable but mutation rules stay enforced.

Required tests:

multi-booking timeline, topic changes, archived topic, archived space, old null-space booking, pagination, rebook preserving space, explicit new-space choice, nonmember denial, mobile view and existing history regression.

8 K2 Contribution evidence

Model 5.6 Terra Effort Medium Phụ thuộc K1 I2

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK K2 CONTRIBUTION EVIDENCE

Implement P1-safe aggregates from authoritative records: settled completed sessions, confirmed contributed duration, accepted assignment reviews and existing public rating policy. Decide visibility per field; private resource, note and feedback content is never public.

Do not recompute GP from hours. GP earned comes from TransactionLog if product policy exposes it. Avoid double counting bookings, retried events or revised submissions.

Required tests:

aggregation correctness, no double count, null legacy data, hidden review, archived space, timezone duration, privacy by viewer, pagination and reconciliation with wallet or ledger.

Keep the UI modest and evidence-oriented; do not build certificates, grades or badges.

8 L1 Feature flags and product analytics

Model 5.6 Terra Effort Medium Phụ thuộc C2 K1

Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK L1 FEATURE FLAGS AND PRODUCT ANALYTICS

Add a kill switch and pilot allowlist for Learning Hub without breaking normal GiveGot navigation. Define server-validated flag behavior and safe fallback.

Implement analytics events from the master spec: learning_invite_created, learning_invite_accepted, learning_first_value_completed, learning_loop_completed, learning_space_returned and learning_rebooked. Use stable schemas, anonymous or minimum identifiers, dedupe and consent policy. Do not send note, task, feedback or resource content.

Define activation as both members present plus first useful artifact or booking within the approved window. Define week 2 and week 4 retention precisely.

Required tests:

flag off, pilot user on, nonpilot off, event schema, dedupe, no private content, retry, activation boundary and existing analytics or navigation regression.

Add a pilot dashboard query or documented SQL, not a large admin BI product.

9 Prompt audit và pilot

9 M1 Pre pilot adversarial audit

Model 5.5 Effort High Phụ thuộc L1

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK M1 PRE PILOT ADVERSARIAL AUDIT

Perform an independent audit. Do not implement new roadmap features and do not silently fix findings.

Check every P0 LH requirement against code and tests. Trace these complete journeys:

1. existing pair invite to accepted LearningSpace to first value
2. Live booking to recap to accepted settlement
3. Exercise Review task to submission to feedback to revision or acceptance
4. Hybrid prework to scheduled meeting to recap to acceptance
5. cancellation no-show dispute and admin support

Audit authorization, object-level access, signed URLs, Pusher payloads, cron secret, migration compatibility, idempotency, concurrency, notification retry, mobile and accessibility.

Run the full command suite and reconciliation scripts with a disposable database. Produce docs/learning-hub/PRE_PILOT_AUDIT.md with severity, evidence, reproduction, affected requirement, suggested smallest fix and release blocker flag.

Acceptance:

- no claim without file, test or runtime evidence
- every P0 requirement is PASS, FAIL or NOT TESTED
- no code changes except audit artifacts
- recommendation is GO, CONDITIONAL GO or NO GO.

9 N1 Final release audit

Model 5.6 Sol Effort Ultra Phụ thuộc M1 blockers fixed

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK N1 FINAL RELEASE AUDIT

Read PRE_PILOT_AUDIT.md and verify each release blocker was fixed by a dedicated task. Audit the whole repository at the latest checkpoint.

Evaluate product requirement compliance, architecture, backward compatibility, authentication, authorization, storage, LearningSpace consistency, booking, learning modes, fulfillment, GP settlement, transaction idempotency, concurrency, cron, notifications, migration, legacy data, chat, Calendar, Meet, reviews, trust, wallet, admin, responsive behavior, accessibility, observability and test coverage.

Run clean-install verification using package-lock.json, database generation and validation, full test suite, lint, typecheck and production build. Run a clean database migration and a representative legacy-data migration. Run E2E journeys where environment permits; mark unavailable checks honestly.

Do not add features. Only make a trivial documentation correction if it cannot affect production; otherwise report a blocker and stop.

Produce RELEASE_AUDIT.md, RELEASE_CHECKLIST.md and a rollback checklist. Release recommendation requires zero unresolved critical or high security, settlement or migration findings.

9 3 Prompt sửa blocker

Dùng prompt này cho từng finding riêng. Không gom nhiều finding không cùng root cause.

REPAIR ONE VERIFIED LEARNING HUB FINDING

Finding ID: [PASTE ID]

Evidence and reproduction: [PASTE FROM AUDIT]

Required expected behavior: [PASTE]

Allowed files or modules: [LIST]

Do not rely on previous chat. Read MASTER_SPEC, CONTEXT_PACKET, IMPLEMENTATION_STATE and the audit finding. Inspect the latest code.

Reproduce the failure before changing code. Implement the smallest root-cause fix.
Add a regression test that fails before and passes after the fix.
Run the task-specific suite and all regressions named by the finding.
Do not refactor adjacent code or fix other findings.
Update state, test matrix and changelog. End with the Completion Contract status.

9 4 Pilot gate

Gate	Khuyến nghị tối thiểu trước khi mở rộng
Security	Không còn critical hoặc high object-level authorization, secret hoặc private-file finding.
Settlement	Reconciliation bằng không; retry và concurrent cron không tạo duplicate credit.
Migration	Clean DB và representative legacy data đều migrate và rollback plan được diễn tập.
Product	5 đến 10 pair hoàn tất onboarding; ít nhất 5 learning loops hoàn chỉnh.
Differentiation	Có pair thực sự dùng Exercise Review hoặc Hybrid, không chỉ Live.
Reliability	Không có stuck fulfillment không có runbook; provider failure không làm mất domain state.
Measurement	Activation, loop completion, rebook, week 2 và week 4 definitions đã cố định.

Các con số pilot là ngưỡng vận hành khuyến nghị, không phải dữ kiện đã được chứng minh. Owner có thể điều chỉnh và phải ghi thay đổi vào ADR hoặc IMPLEMENTATION_STATE trước khi mở gate.

sau này muốn audit toàn app để deploy lên production thì chạy test thế nào, mấy cái regression test, smoke test gì đó nên test bằng automation hay để chạy bằng test của codex hay phải test manual thế nào, và test bằng cách nào, quy trình ra sao, walkthrough step by step thế nào, có cần tạo file excel gì ko, có cần viết test case, testscript, test suits gì ko, nói chung là tôi muốn biết tất tần tật tất cả những gì cần làm trước khi deploy mọi thứ lên production cho user tràn vào xài thật

cho đến thời điểm prompt C1 được hoàn thành, có thêm bất cứ feature mới nào mà tôi cần phải thêm vào feature inventory, sau này tôi muốn audit toàn app để deploy lên production thì chạy test thế nào, mấy cái regression test, smoke test gì đó nên test bằng automation hay để chạy bằng test của codex hay phải test manual thế nào, và test bằng cách nào, có cần tạo file excel gì ko, có cần viết test case, testscript, test suits gì ko, nói chung là tôi muốn biết tất tần tật tất cả những gì cần làm trước khi deploy mọi thứ lên production cho user tràn vào xài thật, tôi đang có mỗi cái file feature inventory à, bro xem qua, bây giờ cứ chạy một prompt xong mà có tính năng gì mới được implemented, executed thì tôi có thêm vào ko, nếu thêm vào thì thêm như thế nào để đảm bảo sau này có thể test lại toàn bộ app chuẩn bị cho deployment ? hay là phải thêm cái tính năng mới đó vào một cái file excel khác mới được, phải có những file gì, chuẩn bị những gì và prompt Codex thế nào trước khi deploy app lên production

10 Prompt mở rộng có điều kiện

Các prompt trong phần này nằm ngoài P0. Mỗi prompt chỉ được chạy sau khi gate tương ứng được owner xác nhận bằng dữ liệu pilot.

10 P1A Video link learning mode

Model 5.6 Terra Effort High

Gate Có ít nhất ba pair yêu cầu và external video link đã được dùng lặp lại

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK P1A VIDEO LINK LEARNING MODE

Add a VIDEO mode using external HTTPS video links only. Do not add raw upload.

Define required objective, delivered video resource, question or acknowledgement and acceptance. Use deliveredAt and a review window; never settle from a synthetic meeting endTime.

Reuse private resources, activity, notifications and fulfillment services. Restrict supported hosts by safe policy or store user metadata without server-side fetching.

Test allowed and denied URLs, member access, delivery, acceptance, revision or questions, timeout, dispute, duplicate settlement, old modes and mobile UI.

10 P1B Document learning mode

Model 5.6 Terra Effort High

Gate Users repeatedly exchange reading material and request a document workflow

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK P1B DOCUMENT LEARNING MODE

Add DOCUMENT mode on top of private LearningResource, task and notes. Required artifacts are a permitted document or external document link, reading objective, expected learner response and optional provider feedback. Do not build collaborative document editing, PDF annotation or malware scanning in this task. Office upload remains disabled unless scanning is already operational.

Completion and settlement follow deliveredAt plus accepted response, not scheduled time.

Test document access, response requirement, revision, timeout, dispute, unsupported Office upload, legacy modes and private-content boundaries.

10 P1C Async question and answer mode

Model 5.6 Terra Effort High

Gate Pilot shows repeated asynchronous questions that do not fit Exercise Review

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK P1C ASYNC QUESTION AND ANSWER MODE

Add ASYNC_QA with question, answer, clarification thread and learner-resolved state. Use structured domain records; do not reuse unrestricted chat as the source of settlement truth.

Define who may answer, when clarification reopens the item, delivery timestamp, resolve window, dispute and settlement eligibility.

Test unauthorized reader, answer delivery, reopen, resolve, timeout, duplicate resolution, dispute race, notification dedupe and chat regression.

10 P2A AI source consent and provenance

Model 5.6 Sol Effort High

Gate Notes and recap usage is sustained and users request AI assistance

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK P2A AI SOURCE CONSENT AND PROVENANCE

Build only the AI safety and provenance boundary. Do not generate summaries yet.

A member explicitly selects source notes, task records, feedback or transcript segments. The server verifies access to every source at execution time, records consent, source IDs, model metadata, prompt version, creator and retention status. Full chat, arbitrary URL content and private files are excluded unless a later approved parser exists.

Define deletion propagation, opt-out, retry and cost limits. Never expose provider keys.
Test cross-space source injection, deleted source, revoked consent, nonmember, prompt injection content isolation, retry, audit metadata and zero-source request.

10 P2B AI draft recap

Model 5.6 Terra Effort High

Gate P2A passes security review and AI gate is approved

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK P2B AI DRAFT RECAP

Use the existing Gemini integration behind a feature flag to generate an editable DRAFT recap from explicitly selected sources only.

Output must show provenance, generation time and draft status. A human must review and publish; AI cannot mark fulfillment complete, change ratings, settle GP or create authoritative facts. Preserve the original manual notes.

Use bounded input, timeout, retry, cost telemetry and safe error states.

Test grounded-source selection, inaccessible source, hallucination-resistant empty evidence behavior, prompt injection content, edit before publish, opt-out, deletion, provider failure and no side effects on fulfillment or settlement.

10 P3A Managed video provider

Model 5.6 Sol Effort High

Gate Raw video demand justifies provider cost and operations

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK P3A MANAGED VIDEO PROVIDER

Integrate Mux or Cloudflare Stream after an ADR compares cost, upload, webhook, signed playback, retention and regional requirements. Do not build a custom encoder or proxy large video through the app server.

Use direct creator upload, verified idempotent webhook, provider asset lifecycle and signed playback after membership authorization. Record processing, ready, failed and deleted states.

Test forged webhook, replay, wrong-space playback, expired token, abandoned upload, processing failure, deletion, quota and provider outage. Add orphan and stuck-processing runbook procedures.

10 P3B Google Meet artifact consent

Model 5.6 Sol Effort Extra High

Gate Workspace account support scopes consent and privacy review are approved

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK P3B GOOGLE MEET ARTIFACT CONSENT

Implement OAuth scope and consent handling for discovering Meet recordings or transcripts. Do not request broader scopes than required and do not auto-import artifacts without explicit member action and policy approval.

Map a Meet conference to an eligible booking, verify participant and space membership, record consent and provider artifact metadata, and handle unavailable artifacts.

Test revoked OAuth, missing scope, wrong conference, nonmember, artifact not ready, duplicate import, consent withdrawal and existing Calendar or Meet creation regression.

10 P3C Transcript processing and retention

Model 5.6 Sol Effort High

Gate P3B approved and transcript retention policy is signed off

Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.

TASK P3C TRANSCRIPT PROCESSING AND RETENTION

Process approved transcript artifacts asynchronously. Store only required metadata and content under the approved retention and deletion policy. Access remains private and all derived AI use requires the P2A source-consent boundary.

Implement job idempotency, status, retries, dead-letter or support path, deletion propagation and audit. Do not auto-complete fulfillment from transcript availability.

Test duplicate job, partial artifact, revoked consent, deletion, wrong-space access, provider outage, retry exhaustion, privacy export and no settlement side effect.

11 Regression matrix

Flow cũ	Invariant phải giữ	Task nhạy cảm	Bảng chứng tối thiểu
Authentication	Google và credentials login, session user id, suspension policy hoạt động	A1 A2	Auth tests và route smoke
Discover và matching	Search semantic và profile discovery không đổi	C2 D1	Build và targeted UI regression
Mentor profile	Public data và booking CTA không đổi	D1 K2	Profile render test
AvailableSlot	Một slot không bị double book	E1 I1	Concurrent booking test
Booking lifecycle	PENDING CONFIRMED COMPLETED CANCELLED MISSED DISPUTED còn đúng nghĩa	E1 I1 I3	Transition matrix và legacy fixtures
Google Calendar	Event tạo sửa hủy theo flow hiện tại	E1 I3	Mock integration regression
Google Meet	Meeting link creation không bị coi là mọi fulfillment	E1 I1	Live and Hybrid tests
Cancellation	Refund và slot release đúng policy	I1 I2 I3	Cancellation scripts
No show and dispute	Funds freeze và admin resolution giữ nguyên	I1 I2 I3	PA 02 and dispute cases
Wallet and ledger	Số dư khớp TransactionLog, không double credit	I2 I3 K2	Reconciliation script
Public reviews	Submission feedback không làm thay đổi public Review	G3 K2	Review visibility regression
Trust score	Không đổi từ private task feedback	G3 K2	Trust history assertion
Chat	Participant đọc gửi realtime bình thường	A1 A2 H2	Chat route and Pusher tests
Notifications	Legacy notification read state và deep link hoạt động	A2 J1	Notification regression
Admin	Role guard và support flow không mở rộng dữ liệu mặc định	A1 I2	Admin authorization tests
AI quiz and roadmap	Không bị phụ thuộc Learning Hub	D1 L1	Route build smoke
Dashboard and history	Legacy rows null LearningSpace vẫn hiển thị	B1 K1	Legacy data fixture
Cron jobs	Secret, batch, retry và idempotency đúng	A2 I3	Deterministic clock tests

12 Lệnh kiểm thử và xử lý failure

12 1 Test ladder

Mức	Lệnh hoặc loại kiểm thử	Khi chạy
T0	npm run db:generate và npx prisma validate	Mọi schema hoặc Prisma client change
T1	npm run typecheck và targeted node:test qua tsx	Mọi code task

Mức	Lệnh hoặc loại kiểm thử	Khi chạy
T2	Service route integration với mock hoặc disposable database	Domain auth storage settlement
T3	Standalone regression scripts liên quan	Mọi task chậm flow cũ
T4	npm run lint và npm run build	Cuối task UI API hoặc cross-module
T5	Full Learning Hub suite migration matrix E2E reconciliation	Cuối block, pre-pilot và release

Task 01 phải ghi exact script names vào package.json và TEST_MATRIX.md. Các prompt sau dùng tên đã được tạo thay vì tự phát minh command mới.

CANONICAL COMMAND SET AFTER TASK 01

```
npm ci
npm run db:generate
npx prisma validate
npm run typecheck
npm run test:learning-hub
npm run test:learning-hub:integration
npm run test:regression
npm run lint
npm run build
```

Run npm ci only for clean-install gates, not repeatedly in every task.

Database-writing tests must use an explicit disposable DATABASE_URL.

Never print secrets or run destructive fixtures against production.

12 2 Mức test tối thiểu theo task

Task	Mức bắt buộc	Audit trọng tâm
00	Baseline T0 T4	Ghi failure nhưng không sửa
01	T1 và self-test runner	Nonzero exit and isolation
A1 A2	T1 T2 T3 T4	Spoofing object access secret payload
B1	T0 T2 T3 T4	Clean and legacy migration rollback
B2 C1	T1 T2 T3	Concurrency membership token
C2 D1	T1 T3 T4 plus UI	Mobile accessibility legacy routes
E1	T0 T1 T2 T3 T4	Booking slot Calendar Meet
F1 F2	T1 T2 T3 T4	Signed URL MIME quota orphan
G1 G2 G3	T1 T2 T3 T4	State role revision privacy
H1 H2	T1 T2 T3 T4	Optimistic lock dedupe redaction
I1 I2 I3	T0 to T5	Transition race ledger reconciliation
J1	T1 T2 T3 T4	Dedupe failure and private payload
K1 K2	T1 T2 T3 T4	Legacy history aggregate privacy
L1	T1 T2 T3 T4	Flag server enforcement analytics privacy
M1 N1	T0 to T5	Evidence-based release decision

12 3 Failure protocol

1. Reproduce và ghi command, exit code, failing assertion cùng fixture.
2. Xác định failure có tồn tại ở baseline Prompt 00 hay do task mới.

3. Nếu do task mới, task không được PASS. Sửa trong cùng scope hoặc báo BLOCKED.
4. Nếu pre-existing nhưng cần test bắt buộc, mở prompt riêng để sửa baseline; không giấu bằng skip hoặc đổi assertion.
5. Không xóa, weaken, snapshot-update hoặc mock bỏ behavior chỉ để đạt màu xanh.
6. Sau fix chạy targeted test trước; chỉ chạy lại full suite khi targeted pass hoặc task là block gate.
7. Ghi limitation cùng owner, impact và follow-up ID. PASS WITH KNOWN LIMITATION không được dùng cho security, settlement hoặc migration blocker.

12 4 Quy tắc chia nhỏ task

- Tách task nếu đồng thời chạm hai vùng high risk như auth và settlement, migration và UI, hoặc storage và AI.
- Tách task nếu cần hơn một migration độc lập, hơn hai domain aggregates hoặc không thể mô tả rollback trong một đoạn ngắn.
- Tách audit khỏi fix. Audit tạo finding; mỗi finding có prompt sửa riêng và regression test riêng.
- Không để Codex tự triển khai task tiếp theo vì còn quota hoặc context. Stop condition là bắt buộc.
- Một task nên có một outcome có thể demo, một owner của state transition và một tập regression hữu hạn.

13 Checklist vận hành thực tế

13 1 Trước mỗi task

- Task phụ thuộc đã PASS và checkpoint có thể đọc từ checkout hiện tại.
- git status được xem trước; user changes được liệt kê và bảo toàn.
- Model và effort khớp registry; không dùng Ultra nếu không phải N1.
- Chỉ copy một task prompt; không dán lại toàn bộ DOCX vào conversation.
- Codex đọc repo memory và relevant source files thay vì quét lại toàn repo không mục tiêu.
- Environment test không trở production; provider test dùng mock.

13 2 Sau mỗi task

- Đối chiếu acceptance từng dòng và xem diff trước khi tin summary.
- Xác nhận test command thật sự chạy, không chỉ được đề xuất.
- Kiểm tra migration, route contract, error state và backward compatibility nếu có.
- Xác nhận IMPLEMENTATION_STATE và CONTEXT_PACKET đã cập nhật ngắn gọn.
- Stage explicit paths. Tạo commit hoặc giữ cùng checkout theo workflow đã chọn.
- Chỉ chuyển task tiếp theo khi status PASS; limitation phải có follow-up owner.

13 3 Cách mở task Codex mới

1. Chọn đúng GiveGot project và bắt đầu từ branch hoặc working tree chứa checkpoint gần nhất.
2. Chọn model và effort theo registry.
3. Dán nguyên prompt của task tương ứng. Không thêm yêu cầu feature khác vào cuối prompt.
4. Nếu Codex phát hiện conflict lớn, để nó dừng và trả decision request. Trả lời bằng quyết định cụ thể rồi tiếp tục cùng task.
5. Khi task hoàn tất, review diff và test evidence. Sau đó mới tạo task Codex tiếp theo.

13 4 Prompt kiểm tra trạng thái nhanh

```
READ ONLY LEARNING HUB STATUS CHECK
Do not change files.
Read docs/learning-hub/IMPLEMENTATION_STATE.md, CONTEXT_PACKET.md,
TEST_MATRIX.md and git status.
Report:
- latest PASS checkpoint
- current dirty files and whether they belong to the last task
- next dependency-ready task
- unresolved blocker or owner decision
- exact model and effort for the next task
- exact tests that next task must preserve
Do not propose or implement additional features.
```

13 5 Prompt đồng bộ context khi drift

```
CONTEXT RECONCILIATION ONLY
Do not change production code.
Compare MASTER_SPEC, ARCHITECTURE_DECISIONS, IMPLEMENTATION_STATE,
CONTEXT_PACKET, TEST_MATRIX and the latest source code.
Identify stale or contradictory statements with file evidence.
Update repository documentation only after proving the current code state.
Do not reinterpret business policy, rewrite history or mark untested work PASS.
Keep CONTEXT_PACKET at most 200 lines and record the reconciliation in CHANGELOG.
```

13 6 Những yêu cầu nên thêm khi cần

Tình huống	Câu bổ sung vào prompt
Có dirty worktree	Treat all pre-existing changes as user-owned. Do not restore, reset, reformat or stage them.
Chạm payment hoặc GP	Show the ledger invariant and prove idempotency under concurrent calls before coding.
Chạm migration	Test clean and representative legacy databases and document rollback before applying to a shared environment.
Chạm provider	Mock provider in tests, verify webhook signatures and keep domain transaction independent from provider availability.
Chạm privacy	Enumerate actor, object and action authorization for every endpoint and signed URL.
Chạm UI	Test empty loading error mobile keyboard focus and long Vietnamese content states.
Quota thấp	Read only files relevant to this task, run targeted tests first and stop when the Completion Contract is satisfied.
Context nghi ngờ stale	Run Context Reconciliation Only before any feature change.

Phụ lục quyết định sản phẩm phải giữ

- LearningSpace là relationship bền vững của một pair, có primarySkill để định vị và subtopics linh hoạt.
- Không có hard unique constraint pair plus primarySkill; hệ thống gợi ý reuse nhưng user có thể tạo space riêng.
- Role mentor hoặc learner thuộc booking hay task cụ thể, không khóa vĩnh viễn ở membership.
- MVP chỉ gồm Live, Exercise Review và Hybrid. Mode phải thay đổi checklist, artifacts và completion rules.
- Hybrid có meeting theo lịch nhưng prework, deliveredAt, review window và settlement không phụ thuộc riêng endTime.
- BookingStatus và FulfillmentStatus giải quyết hai vấn đề khác nhau.
- Private by default, server session owns identity, signed URLs có TTL ngắn và được cấp sau authorization.

- Build workflow, permissions, fulfillment, reputation và history trong GiveGot; tích hợp storage, meeting, email và video provider.
- AI chỉ tạo draft từ nguồn user chọn, có provenance, consent và human review; không có quyền settlement.
- Mọi rollout dùng feature flag, pilot nhỏ, analytics tối thiểu và rollback runbook.